

**BT503 Environment Biotechnology****PAST PAPERS SOLVED QUESTIONS****1. Feature of phenol that make it pollutant?**

Phenols and some detergents may also mimic the activity of oestrogen.

**2. Name enzyme equipment and solution used in genetic engineering?**

the enzymes, solutions and equipment necessary to perform the procedures; the desired piece of DNA to be transferred; a cloning vector; and the recipient cell. It is also essential to have some means of determining whether or not the transfer has been successful. This is achieved by the use of marker genes.

**3. Importance of anaerobic zone?**

The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

**4. Biological process of phospherial removal activated sludge treatment?**

The process of removing phosphorus by accumulating it with biomass is referred to as enhanced biological phosphorous removal (EBPR) process. The phosphorous removal efficiency for biological systems depends on the phosphorus content of the sludge removed and the efficiency of the solids separation process. (Internet)

When the biological excess sludge is wasted from the system, phosphorus is removed... For higher efficiencies of phosphorus removal, effluent polishing methods can adopted: • addition of coagulants (metallic ions): phosphorus precipitation • effluent filtration: removal of the phosphorus present in the suspended solids in the effluent • combination of the addition of coagulants and filtration. (PPTS)

**5. Photodegradation and name its forms?**

Photodegradation involves the process of oxidation (loss of electrons, increase in oxidation state) and hydrolysis (split of chemical bonds by adding water) ..... Photodegradation does not talk about the ageing or degradation of materials with the use of heat or infrared light.

**6. What are living conditions extremophile give example?**

Three examples of extremophiles are Picrophilus torridus (a thermoacidophile adapted to hot, acidic conditions), Antarctic krill (a psychrophile), and the Pompeii worm (a thermophile).

**7. What are application of biosensor in detection of biocides?**

Biosensors have been extensively applied in clinical, food and environmental areas due to the advantages of fast detection speed, high selectivity and sensitivity (Malhotra and Chaubey, 2003). In this study, the contribution of bioluminescence and luminometer makes a biosensor for detection of water pollution.

## 8. Explain genomic libraries?

A genomic library is a collection of the total genomic DNA from a single organism. The DNA is stored in a population of identical vectors, each containing a different insert of DNA. .... The fragments are then inserted into the vector using DNA ligase.

## 9. Biofilter?

Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants. Common uses include processing waste water, capturing harmful chemicals or silt from surface runoff, and microbotic oxidation of contaminants in air.

## 10. Biodiesel?

Biodiesel is a form of diesel fuel derived from plants or animals and consisting of long -chain fatty acid esters. It is typically made by chemically reacting lipids such as animal fat, soybean oil, or some other vegetable oil with an alcohol, producing a methyl, ethyl or propyl ester.

## 11. Plant associated with fungi?

Mycorrhizae: Associations of fungi with roots of vascular plants

mycorrhizae, mycorrhiza or mycorrhizas) is a symbiotic association between a fungus and a plant. The term mycorrhiza refers to the role of the fungus in the plant's rhizosphere, its root system. Mycorrhizae play important roles in plant nutrition, soil biology and soil chemistry.

## 12. Water system?

water system . a river with all its tributaries. a system, including reservoirs and mains, for storing and supplying water for use by a community.

## 13. How plant can use be reduced pollution?

Plants improve air quality through several mechanisms: they absorb carbon dioxide and release oxygen through photosynthesis, they increase humidity by transpiring water vapor through microscopic leaf pores, and they can passively absorb pollutants on the external surfaces of leaves and on the plant root-soil system.

## 14. How animal work as bioindicator two example?

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment.

Example of a bioindicator is lichens. These plants, which live on surfaces such as trees or rocks or soil, are very sensitive to toxins in the air. Greasewood indicates saline soil

## 15. Define phytoremediation?

Phytoremediation /ˌfaɪtəʊrɪˈmiːdiːʃən/ refers to the technologies that use living plants to clean up soil, air, and water contaminated with hazardous contaminants.

## 16. What are biodegradation?

Biodegradation is the breakdown of organic matter by microorganisms, such as bacteria and fungi.

### 17. Bacteriophage?

In bacteria, the most likely candidates for genetic transfer are plasmids and bacteriophages.

### 18. Biosensor advantage?

Advantages of tissues as biosensors include the following: easier to immobilize compared to cells and organelles. the higher activity and stability from maintain enzymes in natural environment.

### 19. Sources dioxin?

Combustion sources, e.g. municipal waste or medical waste incinerators and private backyard barrel burning.  
□ Metal smelting.

- Refining and process sources.
- Chemical manufacturing sources.
- Natural sources

### 20. Method to get quality crop?

The average quality can be increased by planting seeds into fertilized soil. Increasing Farming Skill also increases the chance to harvest a quality crop. If a Seed Maker turns a higher-quality crop into seeds, the seeds are no more likely to produce higher -quality crops when grown and harvested.

### 21. PCB TNT and subtreated biofilter?

### 22.Phase sludge water in sec wastewater treatment?

Sludge is a semi-solid slurry that can be produced from a range of industrial processes, from water treatment, wastewater treatment or on-site sanitation systems. ... Industrial wastewater treatment plants produce solids that are also referred to as sludge.

### 23.In which filed biotechnology used?

• manufacturing process, • waste management, • pollution control

### 24. Thermophile and its used in PCR? Thermophile and application taq polymerase in PCR?

In PCR: DNA Polymerase of *Thermus aquaticus* (Taq Polymerase), *Pyrococcus furiosus* (Pfu Polymerase).  
Clean energy: bioethanol production, hydrogen production fuel cells, etc.

### 25. Define Extremophiles and name three categories?

Extremophiles are animals that live and thrive under extreme environmental conditions

Psychrophiles can tolerate these circumstances.

Halophiles... survive intensely saline environments, such as exist in natural salt lakes or salt evaporation ponds. Acidophiles thrive in the conditions of low pH, typically below 5

## 26. Extremophile in mature?

## 27. Trickling filter?

A trickling filter is a type of wastewater treatment system. It consists of a fixed bed of rocks, coke, gravel, slag, polyurethane foam, sphagnum peat moss, ceramic, or plastic media over which sewage or other wastewater flows downward and causes a layer of microbial slime (biofilm) to grow, covering the bed of media.

## 28. Advantage and disadvantage of incineration?

Advantages: • Drastic volume reduction, • Sterilisation

Disadvantages: • High costs, • Ash disposal, • Atmospheric pollution.

## 29. Effect of dioxin on human?

dioxins may result in skin lesions, such as chloracne and patchy darkening of the skin, and altered liver function. Longterm exposure is linked to impairment of the immune system, the developing nervous system, the endocrine system and reproductive functions

### Long 5 Marks

## 30. Explain environmental factor bioremediation?

bioremediation is the use of living organisms, primarily microorganisms, to degrade the environmental contaminants into less toxic forms. It uses naturally occurring bacteria and fungi or plants to degrade or detoxify substances hazardous to human health and/or the environment.

## 31. Detail note on BCD and use of biosensor to detect these? ??

## 32. Biodiesel advantage?

Biodiesel is a domestically produced, clean-burning, renewable substitute for petroleum diesel. Using biodiesel as a vehicle fuel increases energy security, improves air quality and the environment, and provides safety benefits.

## 33. Solid waste step?

1. Identify the types of waste.
2. Identify the sources of waste.
3. Determine the potential health hazards from waste.
4. Determine the volume of waste generated
5. Identify safe collection method/s.
6. Identify safe transportation method/s.
7. Identify safe disposal method/s.

## 34. Genetic modification in popular plant used to removal mercury? Describe how?

Genetic modification of a poplar to enable mercury to be removed from the soil and converted to a form able to be released to the atmosphere.

mer A gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reductase, which converts mercury from an ionic to a volatile form.

Initially the gene was transferred to *Arabidopsis thaliana*. The gene was transferred by gene guns to poplar tree (*Liriodendron tulipifera*) embryogenic material. The resulting poplar plantlets were found to exhibit tolerance to mercury and to volatilise it at 10 times the rate observed in untransformed yellow poplar plantlets.

### 35. Factor affect bioremediation?

The factors that directly impact on bioremediation are energy sources (electron donors), electron acceptors, nutrients, pH, temperature, and inhibitory substrates or metabolites. One of the primary distinctions between surface soils, vadose zone soils and groundwater sediments is the content of organic material.

### 36.Properties of soil which has effect on pollution?

The application of waste or polluted water into the soil alters its physical and chemical, thereby affecting the growth of agricultural crops and other living organisms. Even the extensive agricultural practices such as fertilizer and pesticide application deteriorate the soil quality.

### 37.Detail note on environmental biotechnology?

Environmental biotechnology. Environmental biotechnology is the branch of biotechnology that addresses environmental problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes.

### 38. Explain recombinant microorganism especially yeast and microorganisms?

GE of microorganisms for use in environmental biotechnology: expansion of metabolic pathways either to modify the existent metabolic capability, or to introduce new pathways.

Many examples... A strain of *Escherichia coli* into which was engineered some 15 genes originating from *Pseudomonas*. These were introduced to construct a pathway able to produce indigo for the dyeing of denim (Bialy, 1997). The traditional method requires the use of toxic chemicals.

## Long 10 Marks

### 39.Briefly describe category of solid waste?

Different categories of solid waste include:

- Non-combustibles: Metal, tin cans, bottles, stones, etc.
- Ashes/dust: Residue from fires used for cooking.
- Bulky waste: Tree branches, tyres, etc.
- Dead animals: Carcasses of domestic animals and livestock
- Hazardous waste: Oil, battery acid, medical waste.
- Construction waste: Roofing, rubble, broken concrete, etc.

### 40. Genomic Library?

Genomic libraries Collection of total Genomic DNA from an organisms. The genomic DNA is isolated, purified and cut up into pieces of a size suitable to be inserted into a cloning vector, following by ligation and transformation into a host cell. These DNA pieces may either be ligated as a total mixture, into a suitable vector to produce a genomic library, or a specific piece may be isolated and prepared as described ago. Genomic libraries are very useful, as they may be amplified, and accessed almost limitlessly, to look for a specific DNA Sequence. If the genomic library is of a eukaryotic origin the genes will contain non-coding regions, called introns. (Prokaryotes do not contain introns). This is a problem if the gene is to be expressed. This problem can be avoided by using cDNA.

#### 41. Sludge treatment procedure?

Sewage sludge treatment describes the processes used to manage and dispose of sewage sludge produced during sewage treatment. Sludge is mostly water with lesser amounts of solid material removed from liquid sewage. ...

Reducing sludge volume may increase the concentration of some of these toxic chemicals in the sludge.

#### 42. Phytovolatilization?

Phytovolatilization is a process, in which plants take up contaminants from soil and release them as volatile form into the atmosphere through transpiration. The process occurs as growing plants absorb water and organic contaminants.

#### 43. What is cloning factor? Why important in genetic engineering?

Cloning allows for the creation of multiple copies of genes, expression of genes, and study of specific genes. ...

Plasmids have been highly engineered as vectors for molecular cloning and for the subsequent large-scale production of important molecules, such as insulin.

#### 44. Southern blotting and western blotting?

In Southern blotting, DNA fragments are resolved by electrophoresis on a gel. These resolved fragments are then shifted to a membrane (nylon or nitrocellulose).

Western blotting: Proteins (e.g., antibodies antigens interaction)

#### 45. Advantages and disadvantages of incineration (2) Incineration:

Advantages:

- Drastic volume reduction, •Sterilization.

Disadvantages:

- High costs,
- Ash disposal,
- Atmospheric pollution.

#### 46. Manipulation of bacteria without genetic engineering (3)

A general procedure is to take a sample of bacteria from, at, or near, the site of contamination from which a pure culture is obtained in the laboratory and identified, using standard microbiology techniques. The 'training' may be required either to improve the bacterium's tolerance to the pollutant or to increase the capabilities of pathways already existing in the bacterium to include the ability to degrade the pollutant, or a combination of both.

#### 47. 3 phases present in submerged bio filter (3)

Typically, a biofilter has two or three phases, depending on the feeding strategy (percolating or submerged biofilter): a solid phase (media); a liquid phase (water); a gaseous phase (air).

#### 48. Co-metabolism

Co-metabolism is defined as the simultaneous degradation of two compounds, in which the degradation of the second compound (the secondary substrate) depends on the presence of the first compound (the primary substrate).

#### 49. Three forces are in the global environment market

A number of forces over which it has little or no control affect a company's marketing activities. Taken together, they make up its external marketing environment, which includes regulatory and political activity, economic conditions, competitive forces, changes in technology, and social and cultural influences.

#### 50. Phytodegradation

Phytodegradation which is also known as phyto-transformation is the breakdown of contaminants taken up by plants through metabolic processes within the plant, or the breakdown of contaminants surrounding the plant through the effect of enzymes produced by the plants.

#### 51. Phytoremediation

The treatments of pollutants or waste (as in contaminated soil or groundwater) by the use of green plants that remove, degrade, or stabilize the undesirable substances (such as toxic metals).

#### 52. Three sources of dioxin production

According to the most recent US EPA data, the major sources of dioxins are broadly in the following types: Combustion sources, e.g. municipal waste or medical waste incinerators and private backyard barrel burning. Metal smelting. Refining and process sources.

#### 53. Conjugation transduction plasmid difference

Conjugation is the transfer of circular DNA called plasmids through cell to cell contact. Transformation is the uptake of 'free' DNA from the environment. Transduction is the transfer of DNA by bacteria-specific viruses called bacteriophage.

#### 54. Biodiesel

Biodiesel is a form of diesel fuel derived from plants or animals and consisting of long-chain fatty acid esters. It is typically made by chemically reacting lipids such as animal fat (tallow), soybean oil, or some other vegetable oil with an alcohol, producing a methyl, ethyl or propyl ester.

### 2 Marks Questions

#### 55. What is activated sludge? (2)

The activated sludge process is a type of wastewater treatment process for treating sewage or industrial wastewaters using aeration and a biological floc composed of bacteria and protozoa.

#### 56. Mechanism of wastewater? (2)

Wastewater treatment is the process of converting wastewater – water that is no longer needed or is no longer suitable for use – into bilge water that can be discharged back into the environment. It's formed by a number of activities including bathing, washing, using the toilet, and rainwater runoff.

#### 57. Diagnosis of mercury intoxication? (2)

A diagnosis we put when we have special clinical picture and information of anamnesis, which specify on a contact with mercury organic connections. The important diagnostic sign of intoxication is a presence of mercury in blood, urine, and at heavy intoxications – in a cerebrospinal liquid.

## 58. Thermophiles and its uses in PCR? (2)

Thermophiles are organisms that grow and thrive at temperatures (60–80°C) that are often too high for mesophiles— most thermophiles are Archaea. *Thermus aquaticus* is a thermophilic microorganism from which polymerase has been isolated and used for PCR (Taq-polymerase). DNA polymerases from other thermophiles are also being used in PCR.

## 59. Taq polymerase use in PCR?

Taq polymerase is a thermostable DNA polymerase I named after the thermophilic bacterium *Thermus aquaticus* from which it was originally isolated by Chien et al. in 1976. ... It is frequently used in the polymerase chain reaction (PCR), a method for greatly amplifying the quantity of short segments of DNA.

## 60. Extremophiles and their categories in nature?

A variety of microbes inhabit extreme environments.

Categories:

1. Psychrophiles
2. Thermophiles
3. Hyperthermophiles
4. Halophiles
5. Xerophiles
6. Alkaliphiles
7. Acidophiles

## 61. Name enzymes, solution and equipment's used in genetic engineering (2) Enzymes:

Restriction endonuclease or restriction enzymes, DNA ligase, DNA polymerases and reverse transcriptase  
Equipment's:

Gel Electrophoresis, plasmids, transformation or transduction, PCR, Identifying transgenic organisms

3 Marks Questions

## 62. What is wastewater and enlist its level?

Wastewater is "used water from any combination of domestic, industrial, commercial or agricultural activities, surface runoff or storm water, and any sewer inflow or sewer infiltration". Wastewater treatment levels

Primary treatment

Primary treatment includes screens, sedimentation and grit removal.

Secondary treatment

Secondary treatment process removes carbonaceous organic matter and nutrients (nitrogen and phosphorus) from wastewater

Tertiary treatment

tertiary treatment includes filtering, disinfecting and preparing wastewater for recycling.

### 63.What is antibiotic and what is antibiotic resistance? 3

An antibiotic is a type of antimicrobial substance active against bacteria and is the most important type of antibacterial agent for fighting bacterial infections.

Antimicrobial resistance (AMR or AR) is the ability of a microbe to resist the effects of medication that once could successfully treat the microbe. The term antibiotic resistance (AR or ABR) is a subset of AMR, as it applies only to [bacteria](#) becoming resistant to antibiotics . (Internet)

### 64. What is microbial pesticides name microbial biopesticides?

Pesticides based on microorganisms and their products have proven to be highly effective, species specific and ecofriendly in nature, leading to their adoption in pest management strategies around the world.

Name: Biofungicides, bioherbicides, bioinsecticides (Internet)

### 65.What is the role of hormones in body and young children?

Hormones are the body's little messengers, which get produced in one part of the body, such as the thyroid, adrenal or pituitary gland, pass into the bloodstream or other body fluid and go to distant organs and tissues where they act to modify structures and functions.

### 66.Thermophiles and application of TAQ polymerase in PCR?

Thermophiles:

Thermophiles are organisms that grow and thrive at temperatures (60–80°C) that are often too high for mesophiles— most thermophiles are Archaea.

Taq polymerase is a thermostable DNA polymerase I named after the thermophilic bacterium *Thermus aquaticus* from which it was originally isolated by Chien et al. in 1976. ... It is frequently used in the polymerase chain reaction (PCR), a method for greatly amplifying the quantity of short segments of DNA.

### 67.Wastewater treatment definition?

Wastewater treatment is the process of converting wastewater – water that is no longer needed or is no longer suitable for use – into bilge water that can be discharged back into the environment. It's formed by a number of activities including bathing, washing, using the toilet, and rainwater runoff.

### 68. How microbes degrade polycyclic aromatic hydrocarbons?

Various bacteria have been found to degrade PAHs, in which degradation of naphthalene and phenanthrene has been most widely studied. ... Principally, bacteria favor aerobic conditions for degradation of PAHs via oxygenase-mediated metabolism

### 69.Bacteria manipulation without genetic engineering in biotechnology?

A general procedure is to take a sample of bacteria from, at, or near, the site of contamination from which a pure culture is obtained in the laboratory and identified, using standard microbiology techniques. Improve the bacterium's tolerance to the pollutant or to increase the capabilities of its pathways. Tolerance may be increased by culturing in growth medium containing increasing concentrations of the pollutant.

#### 5 Marks Questions

### 70.Recombinant microorganisms, yeast and bacteria 5 marks

Recombinant bacteria is a type of bacteria which have the genetic material DNA of the another organism. Recombinant DNA is a technology which is developed by scientists that made it possible to insert a gene into

the genetic material of a bacterium. This “recombinant” micro-organism can now produce such type of protein that coded by the gene. The scientists made the human insulin gene in the laboratory by using this technology.

#### Uses

The Recombinant bacteria is used in different ways to produce different products which are given below:

- Recombinant chymosin
- Recombinant human insulin
- Recombinant human growth hormone

#### Recombinant yeast:

The yeast in which introduce the characteristics according to our desire is called the recombinant yeast. Yeasts are unicellular eukaryotic microbial host cells that offer unique advantages in producing biopharmaceutical proteins.

### 71. Detail note on environmental biotechnology?

Environmental biotechnology is the branch of biotechnology that addresses environmental problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes.

Environmental Biotechnology deals with apparently less dramatic topics, though their importance may be every bit as great, and thus their direct relevance is far less appreciated by bulk of the population.

The goals of Environmental Biotechnology are in everybody's interest... But for most it is simply addressing a problem that could have avoided already.

### 72.What are dioxins and write their sources?

Dioxins are a group of chemically-related compounds that are persistent environmental pollutants (POPs). Dioxins are found throughout the world in the environment and they accumulate in the food chain, mainly in the fatty tissue of animals.

#### Sources:

Dioxins are mainly by-products of industrial processes but can also result from natural processes, such as volcanic eruptions and forest fires. Dioxins are unwanted by-products of a wide range of manufacturing processes including smelting, chlorine bleaching of paper pulp and the manufacturing of some herbicides and pesticides.

### 73.What is surfactants and what is its purpose?

Surface-active agents /surfactants molecules which are active at interfaces and possess both polar(hydrophilic) and nonpolar(hydrophobic)characteristics in the same molecule. Surfactants lower the surface tension (or interfacial tension) between two liquids or between a liquid and a solid.

When surfactants are dissolved in water they orientate at the surface so that the hydrophobic regions are removed from the aqueous environment.

#### Purpose:

Surfactant reduces the surface tension of fluid in the lungs and helps make the small air sacs in the lungs (alveoli) more stable. This keeps them from collapsing when an individual exhales. In preparation for breathing air, fetuses begin making surfactant while still in the womb.

### 74. Write a note on zooplankton? 5

Zooplankton are heterotrophic plankton. Plankton are organisms drifting in oceans, seas, and bodies of fresh water. The word "zooplankton" is derived from the Greek zoon, meaning "animal", and planktos, meaning "wanderer" or "drifter".

Ecology:

Zooplankton is a categorization spanning a range of organism sizes including small protozoans and large metazoans. It includes holoplanktonic organisms whose complete life cycle lies within the plankton, as well as meroplanktonic organisms that spend part of their lives in the plankton before graduating to either the nekton or a sessile, benthic existence.

Nutritional modes in zooplankton

Herbivores: feed primarily on phytoplankton

Carnivores: feed primarily on other zooplankton (animals)

Detritivores: feed primarily on dead organic matter (detritus)

Omnivores: feed on mixed diet of plants and animals and detritus

## 10 Marks Questions

### 75.What is symbiosis and symbiotic nitrogen fixation? 10 Symbiosis:

interaction between two different organisms living in close physical association, typically to the advantage of both. Nitrogen fixation is a process by which nitrogen in the air is converted into ammonia (NH<sub>3</sub>) or related nitrogenous compounds Atmospheric nitrogen, which is molecular dinitrogen (N<sub>2</sub>), is relatively nonreactive and is metabolically useless to all but a few microorganisms The fixation process converts N<sub>2</sub> into ammonia, which is metabolized by most organisms Nitrogen fixation is essential to life because fixed inorganic nitrogen compounds are required for the biosynthesis of all nitrogen-containing compounds As part of the nitrogen cycle, it is essential for agriculture and the manufacture of fertilizer. It is also, indirectly, relevant to the manufacture of all chemical compounds that contain nitrogen, which includes explosives, most pharmaceuticals, and dyes Symbiotic nitrogen fixation occurs in plants that harbor nitrogenfixing bacteria within their tissues. The best-studied example is the association between legumes and bacteria in the genus *Rhizobium* Each of these is able to survive independently (soil nitrates must then be available to the legume), but life together is clearly beneficial to both. Only together can nitrogen fixation take place.A symbiotic relationship in which both partners benefits is called mutualism.

### 76.What is cloning vector? Why it is important in genetic engineering? (10)

A cloning vector is frequently a plasmid or a bacteriophage (bacterial virus) Must be fairly small and fully sequenced, able to replicate itself when reintroduced into a host cell, thus producing large amounts of the recombinant DNA for further manipulation. Also it must carry on it 'selector marker' genes, which are indicators of genomic integrity and activity. A common design of a cloning vector is one which carries two genes coding for antibiotic resistance. The 'foreign' gene is inserted within one of the genes so that it is no longer functional therefore it is possible to discriminate by standard microbiology techniques, which bacteria are carrying plasmids containing recombinant DNA and which are not. To make the manipulations easier, selector marker genes normally contain a multi cloning site (MCS) which is a cluster of sites for restriction enzymes. Insertion of foreign DNA in vector at MSC site causes disruption of these marker genes resulting in loss of the function of that gene. For example, if that gene codes for antibiotic resistance, it will no longer be functional and will not protect the bacterium from that antibiotic.

Viruses may be used as vectors i.e., the DNA of viruses can be used to carry foreign DNA. Most such recombinant viral DNA vectors need phage particles to perform proper gene transfer i.e. transduction. Transfer of naked viral recombinant vector to host cells is called transfection, and tends to have lower uptake rates as compared to transduction.

### 77.Explain in detail the principles of transformation of the plants? 10

transformation is the geneticalteration of a cell resulting from the direct uptake and incorporation of exogenous genetic material from its surroundings through the cell membrane.

Principle:

There are number of principles which are used for the transformation of the plants which are given below:

Vector method:

There are two vector methods which are given below:

Agrobacterium

Viral transformation

Vector Free method:

There are different vector free methods which are given below:

Gene gun

Electroporation

Vector method

There are two vector methods which are given below:

Agrobacterium

Viral transformation Agrobacterium:

Agrobacterium mediated transformation is the easiest and most simple plant transformation. Plant tissue (often leaves) are cut into small pieces, e.g. 10x10mm, and soaked for 10 minutes in a fluid containing suspended Agrobacterium. The bacteria will attach to many of the plant cells exposed by the cut. The plant cells secrete wound-related phenolic compounds which in turn act to up regulate the virulence operon of the Agrobacterium.

Vector method

Viral Transformation:

Package the desired genetic material into a suitable plant virus and allow this modified virus to infect the plant. If the genetic material is DNA, it can recombine with the chromosomes to produce transformant cells. However, genomes of most plant viruses consist of single stranded RNA which replicates in the cytoplasm of infected cell. For such genomes this method is a form of transfection and not a real transformation, since the inserted genes never reach the nucleus of the cell and do not integrate into the host genome. The progeny of the infected plants is virus-free and also free of the inserted gene.

Vector less method

There are two vector less method which are given below:

Gene gun Electroporation

Gene Gun:

It is referred as particle bombardment or biolistics. Particles of gold or tungsten are coated with the DNA and then shot into young plant cells.

Electroporation:

Formation of transient holes in cell membranes using electric pulses of high field strength this allows DNA to enter as described above for bacteria.

## 78. What is Recombinant?

Recombinant is the anything which have the mixture of two hereditary materials one is of the organism and the second is of the another which hereditary material is introduced into the body of first.

### 79. Advantages of Yeast in Recombinant?

Yeasts have been used for thousands of years in food and fermentation processes to produce alcoholic beverages and breads. In recent times, the industrial importance of yeasts has extended beyond its traditional use in fermentation into various healthcare sectors, such as in the production of therapeutic recombinant proteins. Yeast cells are particularly advantageous as hosts for biopharmaceutical production in that they are generally recognized as safe (GRAS) organisms.

### 80. Types of pollution?

There are different types of the pollution which are given below: Air pollution

Water pollution

Land pollution

Environmental pollution

Light pollution

Noise pollution

### 81. Mitochondrial disease & diabetes mellitus

Mitochondrial diseases are long-term, genetic, often inherited disorders that occur when mitochondria fail to produce enough energy for the body to function properly.

Diabetes mellitus (DM), commonly known as diabetes, is a group of metabolic disorders characterized by high blood sugar levels over a prolonged period. Symptoms of high blood sugar include frequent urination, increased thirst, and increased hunger. (Internet)

### 82. Transformation Plant?

Plant transformation is a scientific approach whereby DNA from any organism is inserted into the genome of a species of interest. The inserted DNA is called a “transgene”, and the resulting plant is said to be “transgenic”. Transgenic plants are vital for both research and agriculture.

### 83. Bioaccumulation ?

The collection of substances e.g. toxins and chemicals, in an organism's body is called bioaccumulation.

Sources: substances from any source e.g. air, water, soil, could be stocked through bioaccumulation.

Substances to be gathered through bioaccumulation have following properties;

They are preferably relentless, lipophilic and stable

Unable to translocate throughout the body

Have low catabolic rate

High affinity for proteins

Pollutants take entry to the organism's body or tissue through inhalation, ingestion, epithelial contact or by additional means.

### 84. What is toxicity?

Toxicity is the degree to which a chemical substance or a particular mixture of substances can damage an organism. Toxicity can refer to the effect on a whole organism, such as an animal, bacterium, or plant, as well

as the effect on a substructure of the organism, such as a cell (cytotoxicity) or an organ such as the liver (hepatotoxicity).

## 85. Use of recombinant yeast?

Yeast in medicine Industries

Pharmaceutical proteins comprise one of the fastest growing groups of molecular medicines and currently play important roles in the treatment of many diseases. The estimated market value of biopharmaceuticals, including recombinant therapeutic proteins, nucleic acid-based products, and engineered cell-based products.

## 86. What is plac z expression vector?

Lac -Z promoter operator is in frame with lac-Z alpha fragment (the NH<sub>3</sub> terminal part of Galactosidase gene).

Multiple cloning sites are found in the border of NH<sub>3</sub> end including ATG sequence.

The presence of such restriction site sequences should not disturb the functional activity of the protein, which complements with the omega fragment of the Lac-Z produced by the bacterial cell as the complement.

If any gene is placed in proper frame in the MCS the protein expressed will be in fused form. The expression of the gene can be regulated by IPTG (Isopropyl thio β-Galactoside).